



## Unit 7: Gravity Test

**Directions:** You must show all steps required to arrive at the correct answer for the problem below, including any diagrams.

1. Endor is a gas giant planet with a mass of  $9 \times 10^{27}$  kg. The Forest Moon orbits Endor and is the sight of a famous battle where a bunch of teddy bear creatures defeat the forces of the Galactic Empire using rocks. The Forest Moon has a mass of  $3 \times 10^{23}$  kg and orbits Endor at an average distance of  $6.2 \times 10^8$  m.

- Calculate the orbital speed of the Forest Moon, starting from one of Newton's Laws.
- Calculate the orbital period of the Forest Moon in hours.

2. Din puts Baby Yoda on a scale and measures his mass to be 7 kg.

- Is this Baby Yoda's inertial or gravitational mass? Justify your answer.
- Baby Yoda is on Tatooine, which has four times the radius and half the density of Earth. Calculate Baby Yoda's weight on the surface of Tatooine.



3. Tatooine is a system with two stars, Star A has a mass of  $M$  and Star B has a mass of  $3M$ . The two stars exert a gravitational force on each other.

- Which star experiences a greater force of gravity?

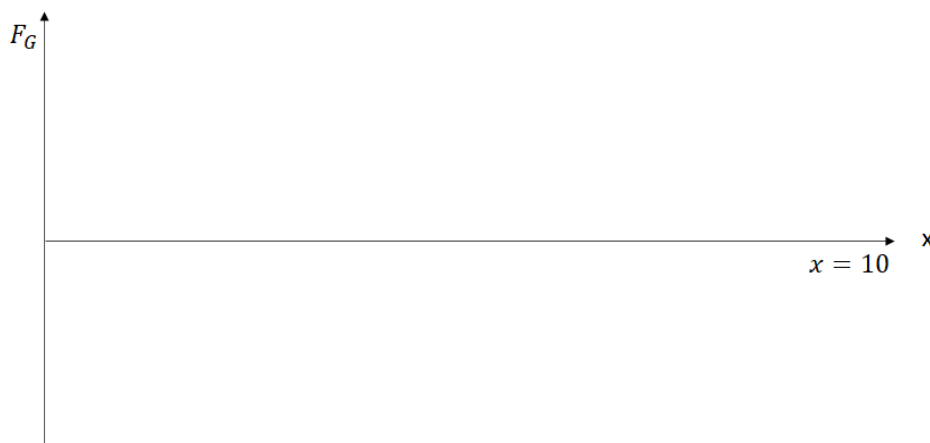
\_\_\_\_\_ Star A

\_\_\_\_\_ Star B

\_\_\_\_\_ The stars experience the same force

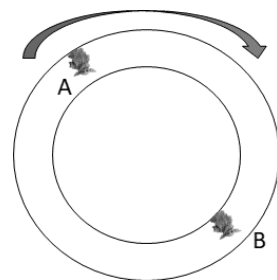
Justify your answer.

- Baby Yoda starts at start A, which is  $x=0$  on a coordinate system. He then travels to Star B. Plot the net force of gravity on Baby Yoda as he travels from A to B, with the right direction being defined as positive.



4. Baby Yoda stands in a spaceship experience artificial gravity. He experiences artificial gravity by standing inside the surface of a solid cylindrical ring as shown.

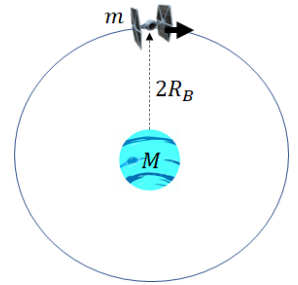
- In what position, A or B, would Baby Yoda be able to stand under acritical gravity. Justify your answer.
- The space station spins at constant velocity. What is the direction of Baby Yoda's acceleration?
- The space station has a radius of 350 m. Calculate the speed at which it needs to spin for Baby Yoda to experience Earth's gravity.



5. Han Solo, a famous smuggler, is from the planet Corellia. Corellia has a mass of  $4 \times 10^{24} \text{ kg}$  and a radius of  $5 \times 10^6 \text{ m}$ .

- Starting from conservation of energy, calculate the escape velocity of Corellia.
- Han Solo takes in the Millennium Falcon from the surface of Corellia. Calculate the minimum speed he must take off with so that he completely escapes Corellia's gravity with a velocity equal to twice the value calculated in a).
- Kinematics could not be used to solve part b). Justify why this is so.

6. A tie fighter of mass  $m = 800 \text{ kg}$  orbits Bespin, a planet where Lando betrayed Han, Leia, and Chewie and handed them over to Darth Vader. Bespin has a mass of  $M = 5 \times 10^{25} \text{ kg}$  and a radius of  $R_B = 5 \times 10^7 \text{ m}$ . The tie fighter orbits at a height of  $R_B$  above the surface of Bespin as shown.



- Calculate the total mechanical energy of the tie fighter as it orbits Bespin. Show your work.
- The energy in a) is negative. Why is this?
- The tie fighter is now moved to a lower orbit by using The Force. Does The Force do positive, negative, or zero work on the Tie Fighter? Justify your answer.

7. Din takes Baby Yoda to Corvus to find the Jedi Ahsoka in the hit streaming series, the Mandalorian. Corvus is a planet with three times the mass of Earth and one half radius of Earth.

- Calculate the acceleration due to Gravity on Corvus.
- This is the way. Din kicks Baby Yoda down an incline. Baby Yoda slides down the incline starting from rest. The incline has a height of  $8 \text{ m}$  and is inclined at  $60^\circ$  with the horizontal as shown. The incline is rough, with coefficient of friction  $\mu = 0.2$ . Calculate Baby Yoda's speed at the bottom of the incline.

